Received: 26 July 2023 | Revised: 8 September 2023 | Accepted: 5 October 2023

DOI: 10.1049/rsn2.12494

IET Radar, Sonar & Navigation

The Institution of Engineering and Technology WILEY

ORIGINAL RESEARCH

A new simulation methodology for generating accurate drone micro-Doppler with experimental validation

Matthew Moore | **Duncan A. Robertson** | **Samiur Rahman**

School of Physics & Astronomy, University of St Andrews, St Andrews, Scotland

Correspondence

Matthew Moore.

Email: mm512@st-andrews.ac.uk

Funding information

Engineering and Physical Sciences Research Council; QinetiQ

Abstract

Unmanned Aerial Vehicles, or drones, pose a significant threat to privacy and security. To understand and assess this threat, classification between different drone models and types is required. One way in which this has been demonstrated experimentally is through this use of micro-Doppler information from radars. Classifiers capable of exploiting differences in micro-Doppler spectra will require large amounts of data but obtaining such data experimentally is expensive and time consuming. The authors present the methodology and results of a drone micro-Doppler simulation framework which uses accurate 3D models of drone components to yield detailed and realistic synthetic micro-Doppler signatures. This is followed by the description of a purpose-built validation radar that has been developed specifically to gather high-fidelity experimental drone micro-Doppler data with which is used to validate the simulation. Detailed comparisons between the experimental and simulated micro-Doppler spectra from three models of drones with differently shaped propellers are given, showing very good agreement. The aim is to introduce the simulation methodology. Validation using single propeller micro-Doppler is provided, although the simulation can be extended to multiple propellers. The simulation framework offers the potential to generate large quantities of realistic drone micro-Doppler signatures for training classification algorithms.

KEYWORDS

micro Doppler, radar

1 | INTRODUCTION

The past decade has seen a significant increase in the number and availabilities of Unmanned Aerial Vehicles, or drones [1]. This increased accessibility has brought benefits to a range of industries, particularly in photography, agriculture and entertainment providing a unique viewpoint at a fraction of the cost of existing technologies. Unfortunately, the accompanying nefarious uses of drones have become an ever-present privacy and safety concern [2–4]. As a result, methods for detecting and classifying drones is a very active area of research. Radar has been shown to be a particularly effective sensor for this, because of its ability to operate in all weather conditions, including at night, and to measure Doppler returns which can be used to distinguish drones from common confusing elements, such as birds [5]. The Doppler return from a target falls into two categories: bulk Doppler and micro-Doppler. The bulk Doppler describes the Doppler shift from the motion of an object as a whole and provides information about the macroscopic motion of a target. Micro-Doppler results from the micro-motions within a target [6]. For a drone, the bulk Doppler relates to the motion of the fuselage of the drone, while the micro-Doppler results from the rotating propellers that most drones use to sustain flight [7].

Drone classification can be considered as two distinct challenges: external and internal classification. External classification describes techniques used to differentiate drones and non-drones, such as birds. Birds are common confusing elements for a radar system, due to their similarities in size, altitude and speed to drones. Consequently, avian targets and drones exhibit comparable bulk Doppler characteristics. However, their micro-Doppler spectra can effectively distinguish between them. The micro-Doppler spectra derived from the motion of bird's wings differs significantly from the propeller micro-Doppler signatures generated by drones. Micro-Doppler has been used to classify between drones and birds experimentally [5, 8–11]. Internal classification attempts to distinguish between different drone types or models. This provides valuable information about the threat posed by a drone and informs users of appropriate countermeasures. Internal classification presents an increased level of difficulty because of the inherent similarities between the micro-Doppler spectra of different drones. However, differences in shape between drone parts (especially propellers) result in unique micro-Doppler spectra when sampled unambiguously in Doppler.

Micro-Doppler information is commonly extracted using a short-time Fourier-transform (STFT), a signal processing technique that extracts spectral components of a signal while preserving temporal information. The STFT presents a trade-off between time and frequency resolution. A long integration time maximises frequency resolution at the expense of the temporal resolution, and vice versa [12]. Using a short integration time produces micro-Doppler information showing the sinusoidal blade-tip motion traced out by the propeller. Another important feature that emerges for short integration times is blade flash. This occurs when the propeller aligns perpendicularly to the radar's line-of-sight causing the return from multiple scattering points along the propeller blade to coherently add up in phase. This coherent addition results in a flash being produced [13]. Blade flash depends on the exact shape of the propeller such that if resolved, the feature forms a signature for the drone [14].

Training appropriate classifiers to exploit micro-Doppler signatures for internal classification will require large datasets encompassing a range of parameters. Generating these datasets experimentally is time-consuming and logistically challenging. A solution to this is to generate the data synthetically. In this paper, a simulation framework capable of generating realistic micro-Doppler data at millimetre-wave frequencies from computer-aided design (CAD) models of drones is presented. This approach has the advantage that it removes the need to gather in-flight data from drones allowing for rapid augmentation of classifier training datasets as new drones become available. The simulation has been validated experimentally and shown to produce synthetic signatures which closely match measured signatures from the same drone. The paper is structured as follows: Section 2 details the prior art in this field including the use of micro-Doppler for drone classification and simulation approaches. Section 3 describes the novel simulation approach presented in this paper including the mathematical basis. Section 4 explains the design and performance of the validation radar and Section 5 compares the simulated and experimental results for three different drone models both qualitatively and quantitatively.

2 | MICRO-DOPPLER FOR DRONE CLASSIFICATION

Differentiating between different models or types of drones (internal classification) can be done by using the information contained within the micro-Doppler return from a drone. Harmanny et al. used micro-Doppler information passed into a maximum a posteriori probability classifier to distinguish between a small fixed-wing drone, a small single propeller drone and a multirotor drone [5]. Bjorklund et al. used micro-Doppler in the form of the Kim-Ling features and a boosted classifier to classify several different drones (and a bird) with 80% accuracy [15]. Rahman et al. used the blade flash and tip-trace information as a signature of a drone to classify between three drones in-flight [14]. Fully resolved Doppler spectrograms generated with fine temporal resolution provide more features with their temporal variations for use by a classifier. With continuous-wave (CW) radars, it is feasible to achieve the required Doppler sampling rate to unambiguously sample the propeller micro-Doppler, and millimetre wave radars offer high Doppler sensitivity in a short coherent integration time to maximise the useful features. Accordingly, this work focusses on radars in the millimetre range (specifically 94 GHz) to exploit blade flash information for classification.

There are several full electromagnetic (EM) solver-based simulations that are capable of generating the radar return from a target. Generally, these methods calculate the scattering from a given geometry and by iterating this geometry they can simulate the micro-Doppler signature of a target, including a drone [6]. These numerical methods include the Method of Moments (MoM), which approximates surface currents, and the Finite Element Method (FEM), which approximates volumetric currents, both of which are relevant at lower frequencies (in the Rayleigh and resonant scatter regimes). Schröder et al. developed a full-body micro-Doppler simulation of a DJI Phantom 2 using MoM at 10 GHz [16]. Speirs et al. used MoM and FEM to generate spectrograms of a DJI Phantom 2 at X-band [17]. Both showed the general shape of the micro-Doppler return, including the blade flashes but lacked the high-fidelity details required to perform internal classification because of the lower frequency used. An alternative approach is to use ray tracing-based simulations which are relevant at higher frequencies such as the millimetre-wave frequencies considered in this paper. One example is Physical Optics (PO) where a target is constructed of shapes of known scattering behaviour. Schröder used PO to generate spectrograms for the DJI Phantom 2 propeller and compared with their MoM simulation [16]. They found good agreement in general shape, but that the Doppler spectrum was not well represented by PO and that these simulation techniques are not appropriate for micro-Doppler investigations. One significant problem associated with numerical simulations is their considerable runtime. Using realistic materials, Schröder et al.'s numerical simulation would have taken 160 days [16] which precludes the use of these for generating large datasets for training classifiers. For such an application, a computationally efficient simulation approach is needed that can generate large datasets on a reasonable timescale.

An alternative approach to full EM solvers with much shorter runtimes is to simulate the micro-Doppler information only, by considering a target as a collection of point scatterers as first suggested by Chen et al. [18]. This approach can be modified to consider objects as thin wires [19]. Melino et al. applied this model to generate spectrograms in good agreement with experimental results at 10 GHz [20]. Cai et al. modelled a propeller as two thin wires and compared the results to experimental results obtained in an anechoic echo chamber, at X-band [21]. The two results demonstrated good agreement at these frequencies, but the simulation lacked the Doppler resolution required to differentiate between models.

3 | MICRO-DOPPLER SIMULATION

For a simulation to be useful for internal classification, it must fulfil three requirements. Firstly, it must generate micro-Doppler data from targets with sufficient detail such that the micro-Doppler differences are present. Secondly, it must be of high accuracy to be relevant to real world classification and finally, it must be able to provide this data relatively quickly and easily, using inputs that are readily available. The requirement for accurate micro-Doppler results that are unique to a certain drone precludes the use of simplified models of drone components. As such, this simulation approach uses point clouds derived from accurate 3D CAD models. This simulation can generate micro-Doppler information from moving drone components, including propellers, with high levels of accuracy and detail. The computation time is significantly lower than alternative simulation approaches and individual points are treated independently such that the runtime can be accelerated by using multithreading.

3.1 | Simulation theory

A target is modelled as a collection of points, as first suggested by Chen et al. [18]. The scattering from a point of index j, is given by Equation (1),

$$S_j(t) = \sigma_j e^{\frac{i4\pi}{\lambda}\bullet\left|\overrightarrow{R_j}(t)\right|}, \tag{1}$$

such that the phase can be calculated for any point in time. σ_j is the radar cross-section (RCS) of the point, λ is the wavelength and $\left|\overrightarrow{R_j}(t)\right|$ is the distance between the point and the radar as a function of time. The RCS of all points has been standardized to one. The scattering from the target as a whole can be approximated for its given position by coherently summing the scattering from the N individual points, as given by Equation (2).

$$S_{target} = \sum_{j=1}^{N} \sigma_j e^{\frac{i4\pi}{\lambda}\bullet\left|\overrightarrow{R_j}(t)\right|} \tag{2}$$

In order to calculate the micro-Doppler returns for a target, the temporal variation of S_{target} must be found. This variation comes from the $\overrightarrow{R_j}(t)$ term, which describes the time dependent position of the individual jth scatterer. By considering the specific motion of a propeller's rotation about its centre, a more general expression for the radar return from a propeller's motion can be found. Consider, without loss of generality, that the propeller's centre of rotation is positioned at the origin and the radar is viewing the propeller from some distance away. There exists a vector from the radar to the static centre of rotation of the propeller (the origin), $\overrightarrow{C}$, and a vector from the centre of rotation to the jth point scatterer on the rotating propeller, $\overrightarrow{P_j}(t)$, whose time dependence is associated with the propeller's rotation. Figure 1 shows these vectors pictorially.

$\overrightarrow{R_j}(t)$ can be written as a sum of $\overrightarrow{C}$ and $\overrightarrow{P_j}(t)$. Furthermore, $\overrightarrow{P_j}(t)$ can be expressed as a matrix multiplication of the rotation matrix $R_z(t)$ and the start position of the jth scatterer, $\overrightarrow{P_{0j}}$.

$$S_{target} = \sum_{j=1}^{N} \sigma_j e^{\frac{i4\pi}{\lambda}\left|\overrightarrow{C}+\overrightarrow{P_j}(t)\right|} \tag{3}$$

$$S_{target} = \sum_{j=1}^{N} \sigma_j e^{\frac{i4\pi}{\lambda}\left|\overrightarrow{C}+\mathrm{R}_z(t)\ \overrightarrow{P_{0j}}\right|} \tag{4}$$

Where $\mathrm{R}_z(t) = \begin{bmatrix} \cos(\theta(t)) & -\sin(\theta(t)) & 0 \\ \sin(\theta(t)) & \cos(\theta(t)) & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $\theta(t)$ is angle of rotation about the z-axis. If non-rotating targets are included in the simulation (such as the drone's fuselage) these M points (indexed by k) can be included as the second term in Equation (5).

$$S_{target} = \sum_{j=1}^{N} \sigma_j e^{\frac{i4\pi}{\lambda}\left|\overrightarrow{C}+\mathrm{R}_z(t)\ \overrightarrow{P_{0j}}\right|} + \sum_{k=1}^{M} \sigma_k e^{\frac{i4\pi}{\lambda}\left|\overrightarrow{P_k}\right|} \tag{5}$$

The simulation was performed for various elevation angles, β, the angle below the plane of the propeller's rotation (the xy plane in Figure 1). This is equivalent to rotating all targets about the y-axis by β, for the propeller's centre of rotation

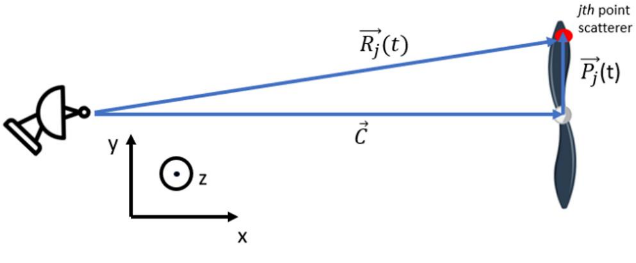

FIGURE 1 Pictorial description of vectors used in simulation.

located at the origin. This leads to a generalised Equation (6) for the micro-Doppler return from a drone's propeller for any elevation angle.

$$S_{target} = \sum_{j=1}^{N} \sigma_j e^{\frac{i4\pi}{\lambda} R_y(\beta) \left| \overrightarrow{C} + \mathrm{R}_z(t)\ \overrightarrow{P_{0j}} \right|} + \sum_{k=1}^{M} \sigma_k e^{\frac{i4\pi}{\lambda} R_y(\beta) \left| \overrightarrow{P_k} \right|} \tag{6}$$

Where $\mathrm{R}_y(\beta) = \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix}$.

The $N + M$ point scatterers used in the simulation are derived from accurate CAD models. However, for a given geometry, certain portions of the propeller will be obscured. Such shadowing effects are incorporated by using Katz's Hidden Point Removal (HPR) algorithm [22]. This algorithm is formed of two steps. First, all points are spherically reflected in a sphere of radius, R, centred at the radar viewpoint as given by Equation (7),

$$\hat{p}_i = F(p_i) = p_i + 2(R - |p_i|) \frac{p_i}{|p_i|} \tag{7}$$

where p_i corresponds to the points prior to performing HPR, $\hat{p}_i$ to points after HPR, F is the spherical flipping operator and R is the radius of the sphere used, a user definable parameter. Points that lie on the convex hull of the transformed points and the viewpoint are visible. The remaining points post HPR, $\hat{p}_i$, form the $N + M$ points used to simulate the radar return of the target in Equation (6).

3.2 | Simulation implementation

In this paper, we have used a frequency of 94 GHz for the high Doppler sensitivity associated with millimetre-wave frequencies and corresponding high-fidelity micro-Doppler returns but the method is applicable at other frequencies. The 3D CAD models were sourced from online libraries before being converted into point clouds using CloudCompare, a free software package used for working with point clouds [23]. In order to accurately represent the drone component's surface, the CAD models were over-sampled by a factor of 10 before being down-sampled at simulation runtime. Neglecting the over-sampling resulted in fine features of the propellers (especially the blade edges) not being accurately represented. Unfortunately, open-source CAD models of drone components were difficult to find in some cases and often lacked accuracy, particularly of the subtly curved surfaces of the propeller blades. In these situations, high resolution laser scanning of propellers was used to create the point clouds, discussed further in section 3.3. The number of points included in the simulation was a compromise between micro-Doppler detail and simulation runtime. Next, shadowing effects were modelled by removing hidden points from the simulation. Depending on the simulated geometry, not all parts of a component are visible at a given point in time. This means that some of the points in the simulation are obscured and should be removed. This was done using the Katz's HPR algorithm [22]. Figure 2 shows the result of this algorithm, for example, orientation where the radar is viewing the propeller in the plane of rotation and from the lower left of the figure. The nearside blade is largely unaffected but the farside blade is obscured behind the central root of the propeller. Consequently, these obscured points are removed, and the shadowing effect is replicated.

The scattering from the remaining points was calculated individually using Equation (2). It is assumed that the entire target (including the fuselage when included) falls within a single antenna beamwidth. Therefore, by coherently summing these individual returns, an approximation for the scattering from the simulated target for a given orientation was found. Because the scattering from each point is calculated independently, it was possible to speed up this process with multi-threading. The drone's propeller was rotated about its centre by a small angle (~0.001 *radians*) and the scattering was calculated again. This process was repeated until the propeller had rotated around some predefined angle (normally a multiple of 2π), replicating the motion of a real propeller. Using the phase information calculated for each angular position, an STFT was used to generate a spectrogram showing the micro-Doppler return for the associated motion. Although an STFT was used throughout this work, this simulation methodology is agnostic to time-frequency transform. The entire workflow of the simulation can be seen in Figure 3.

In this paper, we primarily focus on the micro-Doppler signature of drone propellers due to their distinctive shape, however, where possible, the fuselage of the drone is also included for shadowing purposes. This simulation approach does not produce accurate RCS or polarisation information. The return power from each scattering point is arbitrary (and set to unity for this paper) and therefore, the total power calculated by the simulation is not accurate. Relative scattering effects are included due to the HPR and coherent summation

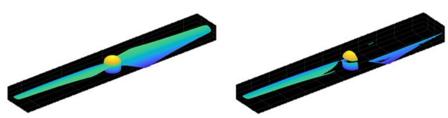

FIGURE 2 Example propeller before (left) and after (right) Katz's Hidden Point Removal (HPR) algorithm removing shadowed points. The radar is viewing the propeller in the plane of rotation from the lower left of the figure.

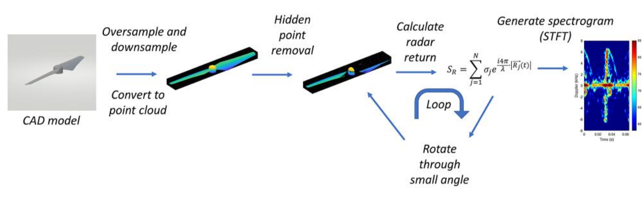

FIGURE 3 Simulation workflow. NB: the first step is skipped if using a point cloud from a laser scanner.

performed by the simulation, but absolutely accurate results require a full EM solver, which this is not. For the same reason, the simulation does not account for polarisation. The authors believe that this is an acceptable compromise for the considerably increased computational efficiency of this approach and the fact that the objective of this work is to develop internal classification tools using micro-Doppler information only. Ultimately, this simulation methodology is able to generate large micro-Doppler datasets in a much shorter timescale than otherwise possible. The approach is valid for targets under far-field conditions, and it ignores multipath effects. These are reasonable assumptions for a drone flying at least several tens of metres from the radar.

Three drones were chosen for this simulation: the DJI Phantom 3 Standard quadcopter, the DJI S900 hexacopter and the Joyance JT5L-404, a crop-spraying quadcopter. These drones were available at the University of St Andrews to perform validation measurements. Furthermore, they vary in size and shape and have significantly different propeller shapes.

FIGURE 4 DJI Phantom 3 (top), DJI S900 (middle) and Joyance JT5L-404 (bottom) with their propellers.

Figure 4 shows each of these drones along with their accompanying propellers.

3.3 | Sourcing CAD models

For the Phantom 3, the CAD model required for simulation was sourced from GrabCAD, an online CAD library [24]. However, for the remaining two drones, accurate CAD models could not be found online so 3D scanning technologies were investigated. These were the Einscan-SE, a camera-based scanning system and a specialised laser scanning service [25, 26]. The camera-based scanning system struggled to capture the fine edges of a drone propeller accurately and was not able to reproduce a smooth propeller surface reliably. Figure 5 shows a CAD model for the Phantom 3 and an associated optical scan for the same propeller.

Figure 6 shows an example spectrogram using the CAD models sourced from the online library and from the camera-based scanner, respectively.

The result using the optical scanner contains significantly more noise when compared to the GrabCAD model result, and the micro-Doppler features are only partially recreated. Consequently, this scanner was not able to provide sufficiently accurate CAD models for the simulation. The specialised laser scanning service used a Nikon H120 handheld laser scanner. This scanner has an accuracy of 7 microns and a point spacing of 0.035 mm. Figure 7 shows an example model for a propeller after laser scanning.

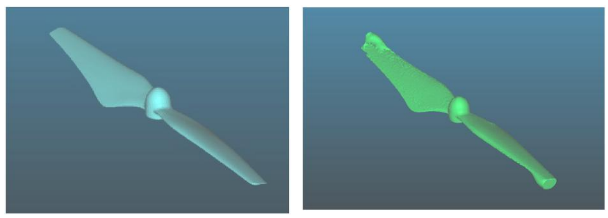

FIGURE 5 DJI Phantom 3 propeller CAD model from GrabCAD (left) and from Einscan-SE scanner (right). CAD, computer-aided design.

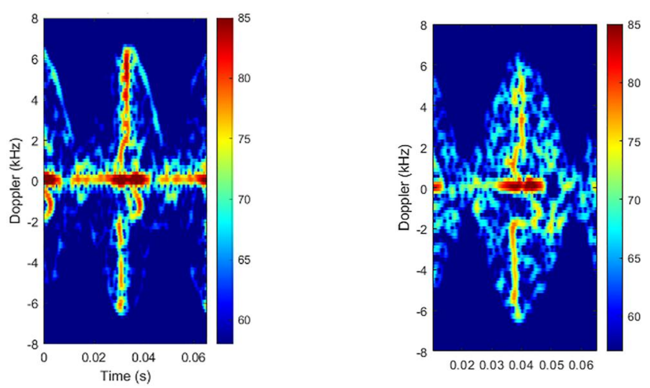

FIGURE 6 Simulated spectrogram using accurate CAD model sourced from online library (left) and CAD model from optical scanner (right). CAD, computer-aided design.

The results were significantly improved, when compared to the optical scanner. The shape, and surface of the propeller were accurately captured to a level that the inscription at the base of the propeller could be read from the scans. Point clouds from the laser scanner were fed directly into the simulation. Starting with point clouds (as opposed to CAD models) had the limitation that the propellers could not be combined with other drone components (motors, fuselage etc.) to incorporate their shadowing effects.

3.4 | Simulation results

For the three drones considered, the radar return from the drone was simulated for a range of elevation angles (β) defined as the angle between the propeller's rotation plane and the radar's line of sight to the drone, as shown in Figure 8. This ensured that different micro-Doppler features associated with different perspectives below the drone were simulated. Furthermore, the maximum extent of the micro-Doppler signature is expected to reduce with elevation angle. The maximum blade-tip Doppler for a propeller viewed at a given elevation angle, β, will correspond to the line-of-sight velocity of the blade-tip, which is given by Equation (8),

$$max\ tip\ Doppler = \frac{4\pi L\dot{\theta}}{2\lambda}\cos(\beta) \tag{8}$$

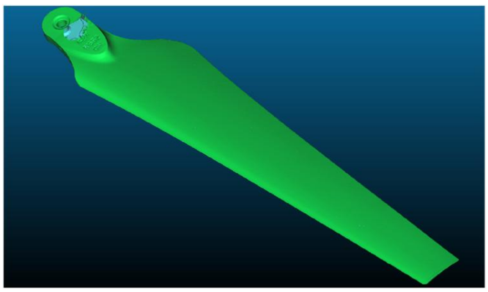

FIGURE 7 DJI S900 model from laser scanner.

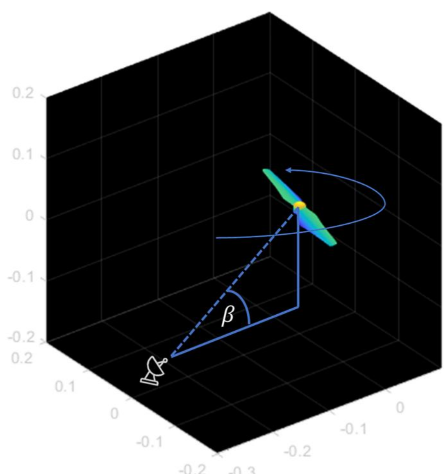

FIGURE 8 Elevation angle definition.

where L is the tip-to-tip propeller length, $\dot{\theta}$ is the rotation frequency and λ is the wavelength. The azimuth angle from which the target is viewed is zero for all simulations. Varying the azimuth angle only changes the start point of the propeller micro-Doppler and the specific shadowing, which is considered to be arbitrary.

The simulation assumed a wavelength of 3.19 mm (94 GHz) and the rotation rate of each propeller was set to its respective idling speed, between 7.5–20.0 Hz. Each spectrogram takes around 10 min to run on a Desktop PC, using 6 threads. It is expected that this can be significantly optimised, and runtime should scale inversely with number of threads.

3.4.1 | DJI Phantom 3

Figure 9 shows the CAD model used for the DJI Phantom 3. The propeller, motor and a portion of the frame were simulated to recreate a real scenario as closely as possible. The radar viewed the propeller from below the plane of rotation of the propeller at six elevation angles ranging from zero to 50° in 10° steps. The frame of the drone is included for its shadowing effect and contribution around zero Doppler.

Figure 10 shows the simulated micro-Doppler returns for the six elevation angles considered. The micro-Doppler associated with the rotating propeller is clearly visible in all plots. The blade flash feature, the maximal reflection of the blade when perpendicular to the line of sight of the radar, can be seen in all plots. At lower elevation angles, the blade flash contains the signature S-shape associated with the DJI Phantom 3 propeller blade. The S-shape collapses as the elevation angle increases and the propeller is viewed further below the plane of its rotation. Furthermore, the micro-Doppler contains a clear sinusoidal trace following the blade-tip's motion. The maximum blade-tip Doppler reduces with elevation angle as the line-of-sight velocity of the blade-tip decreases and the falling edge of the sinusoidal trace is strongest. Another noticeable feature is the asymmetry in the scattering strength in the Doppler axis, especially apparent at higher elevation angles.

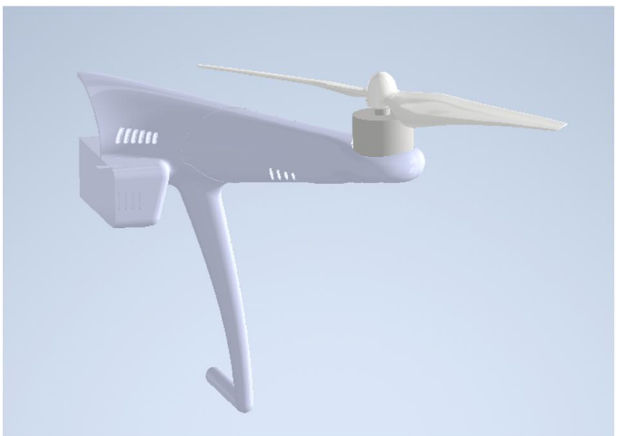

FIGURE 9 CAD model for DJI Phantom 3. CAD, computer-aided design.

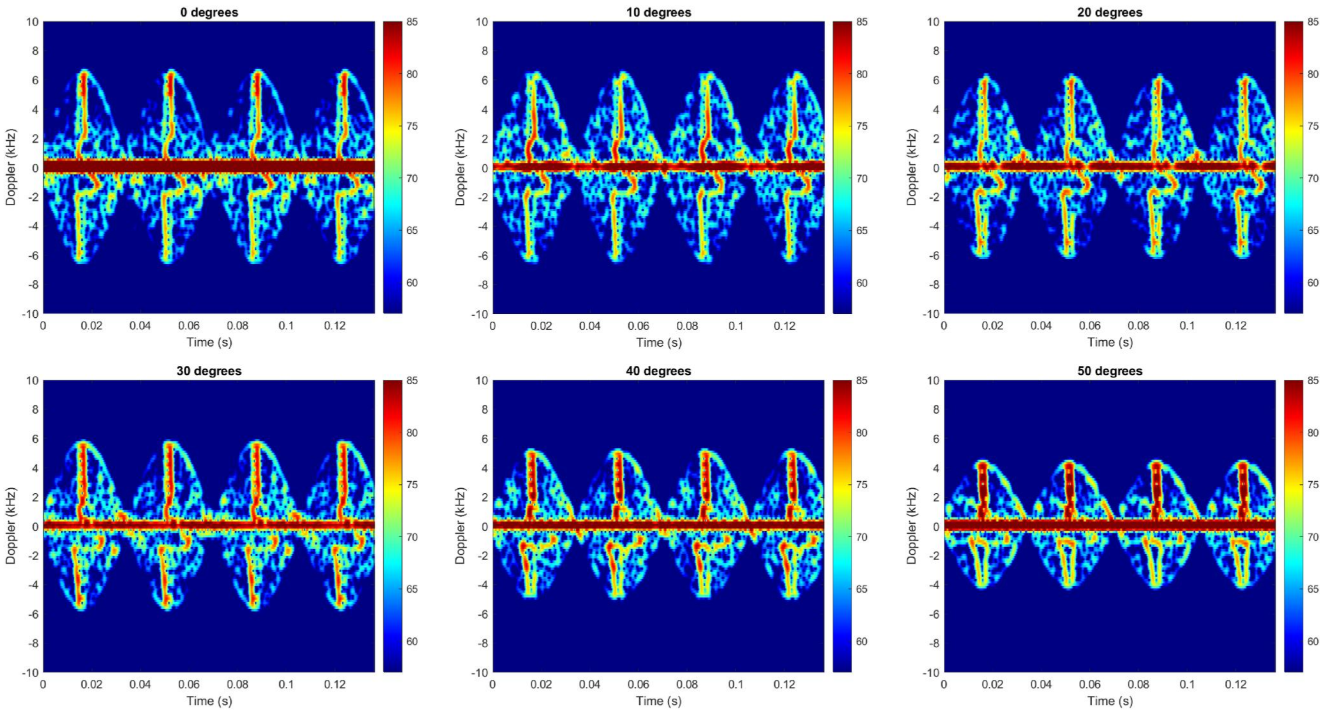

FIGURE 10 Simulated spectrograms showing micro-Doppler for the DJI Phantom 3 for elevation angles ranging from 0 to 50°.

FIGURE 11 DJI S900 (left) and Joyance JT5L-404 (right) propeller point clouds.

The propeller blades are moulded at a certain angle of attack. At a sufficiently large elevation angle below the plane of rotation of the propeller, because of this angle of attack, one propeller blade presents a larger area to the radar than the other. This effect can be seen in the significantly stronger return in the positive Doppler blade flash for elevation angles of 30° and above. Finally, the spectrograms also contain a strong zero Doppler return from the static frame.

3.4.2 | DJI S900

Figure 11 (left) shows the DJI S900 propeller point cloud after laser scanning. Note that the fuselage, clamp-bar (which connects the propeller to the motor) and motor are not included and that only the propeller is simulated.

Figure 12 shows the micro-Doppler return from this propeller for elevation angles from 0 to 50°. In general, the blade flash feature looks quite different from the DJI Phantom 3 blade flash. The blade flash appears as two long thin vertical strips, above and below the zero Doppler horizontal. These vertical strips are not quite symmetrical about the zero Doppler horizontal, with the positive feature shifted slightly to the right, and the negative feature slightly to the left. At lower Doppler frequencies, the blade flash contains a reverse S-shape feature. This bottom of this reverse S links up to the negative portion of the blade flash via the curved section. The oval feature at the base of the blade flash is an artefact from the roots of the propeller. This feature would not exist in reality when these roots are obscured behind clamp bars and motors. The results for all elevation angles contain a sinusoidal pattern representing the blade-tip trace and the falling edge of this sinusoid is strongest. As expected, the blade-tip trace maximum Doppler reduces with elevation angle.

3.4.3 | Joyance JT5L-404

Figure 11 (right) contains the point cloud showing the Joyance JT5L-404 propeller and clamp bar (but not the motor or fuselage) after laser scanning. For this model, the laser-scanned point cloud was ~50% denser in one blade than the other. This resulted in an artificial asymmetry in the micro-Doppler return. To mitigate this, the less dense blade was dropped, and a mirror image of the denser blade used in its place. The

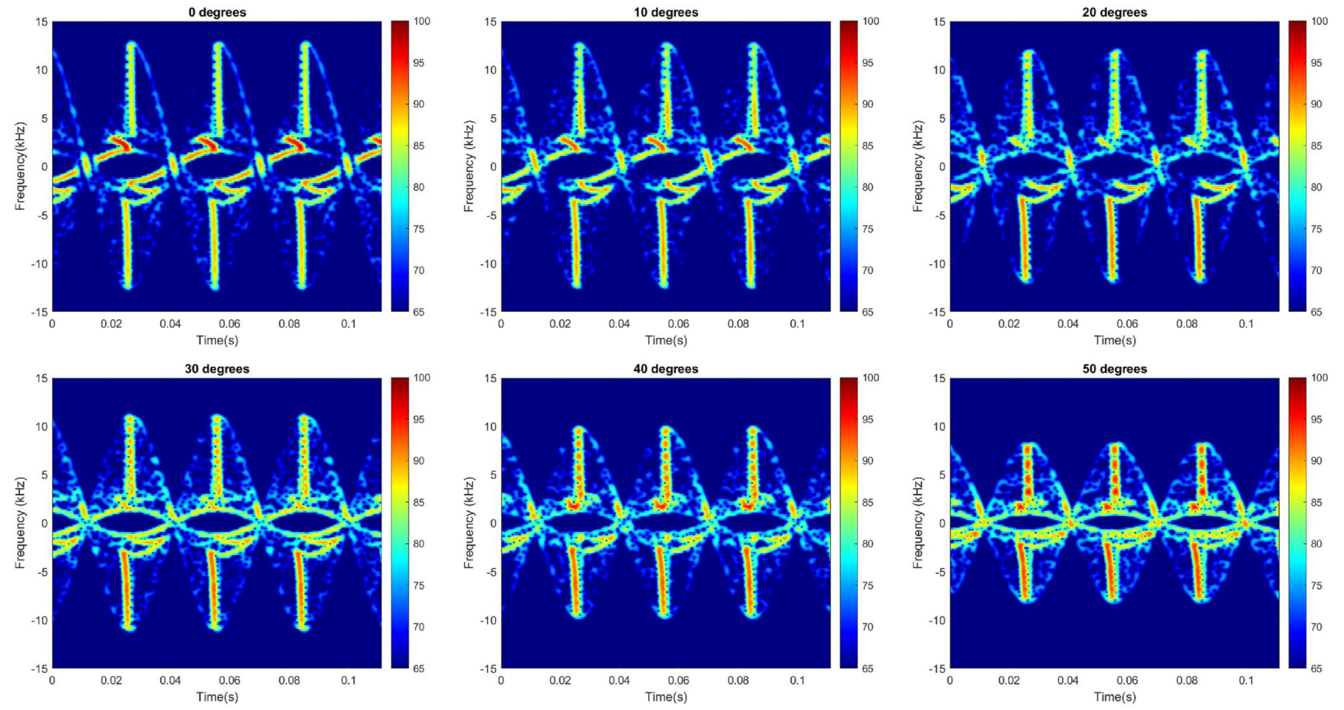

FIGURE 12 Simulated spectrograms showing micro-Doppler for the DJI S900 for elevation angles ranging from 0 to 50°.

resulting micro-Doppler spectrograms are shown in Figure 13. The blade flash for this propeller is clearly visible for all aspects angles. It appears as a long thin feature tapering down to a point in the negative Doppler. Furthermore, the negative portion of the blade flash contains an upside-down L-shape, which is not seen in the positive Doppler. The sinusoidal blade tip trace is visible, the falling edge of which is significantly more prominent than the rest.

4 | VALIDATION RADAR

In order to ensure the simulation's accuracy it was validated experimentally by comparing simulated results with experimental results captured under similar conditions. This process allows for the simulation to be adjusted to reproduce experimentally observed micro-Doppler features and therefore maximise its usefulness. A 94 GHz radar was designed and built for the purpose of validating the simulation. The radar was primarily intended as an indoor radar to be used in the laboratory where simulated scenarios could be exactly recreated. Table 1 shows the key experimental parameters that were considered when designing the radar for this application.

The maximum blade-tip Doppler frequency will occur for a propeller viewed in the plane of rotation ($\beta = 0°$) and can be calculated using the maximum tip-to-tip propeller length, L, the rotation frequency, $\dot{\theta}$, and the wavelength using Equation (8). The required 3 dB antenna beamwidth, θ_{ant}, for a propeller of length, L, to be fully contained within the beam when at a range, R, is given by Equation (9).

$$\theta_{ant} = 2\tan^{-1}\left(\frac{L}{2R}\right) \tag{9}$$

The associated radar requirements are given in Table 2.

The radar operates at 94 GHz and in CW mode in order to match the simulation. CW operation has the advantage that the high sampling rates required to unambiguously sample a rotating propeller are easily obtainable, as well as high Doppler resolutions. The intermediate frequency (IF) signal is sampled at 40 kHz, such that the maximum unambiguous Doppler frequency is 20 kHz, sufficient to fully sample our maximum possible tip Doppler of 19.7 kHz, as shown in Table 2. A 512 bin Fast Fourier Transform (FFT) is used, given the Doppler bin width is 78 Hz, which allows for high resolution micro-Doppler spectrograms of moving propellers. The combination of high sampling rates with good Doppler resolution produces high-fidelity spectrograms. Whilst resolving the maximum blade-tip Doppler is essential, a lot of the useful micro-Doppler information is contained at much lower frequencies. Achieving high quality data as close to 0 Hz as possible allows for better evaluation of simulation accuracy. Consequently, IF amplification and filtering were an important consideration when building this radar.

4.1 | Design

A detailed description of the validation radar has been presented previously [27]. A simplified block diagram is given in Figure 14. For this application, only CW operation was

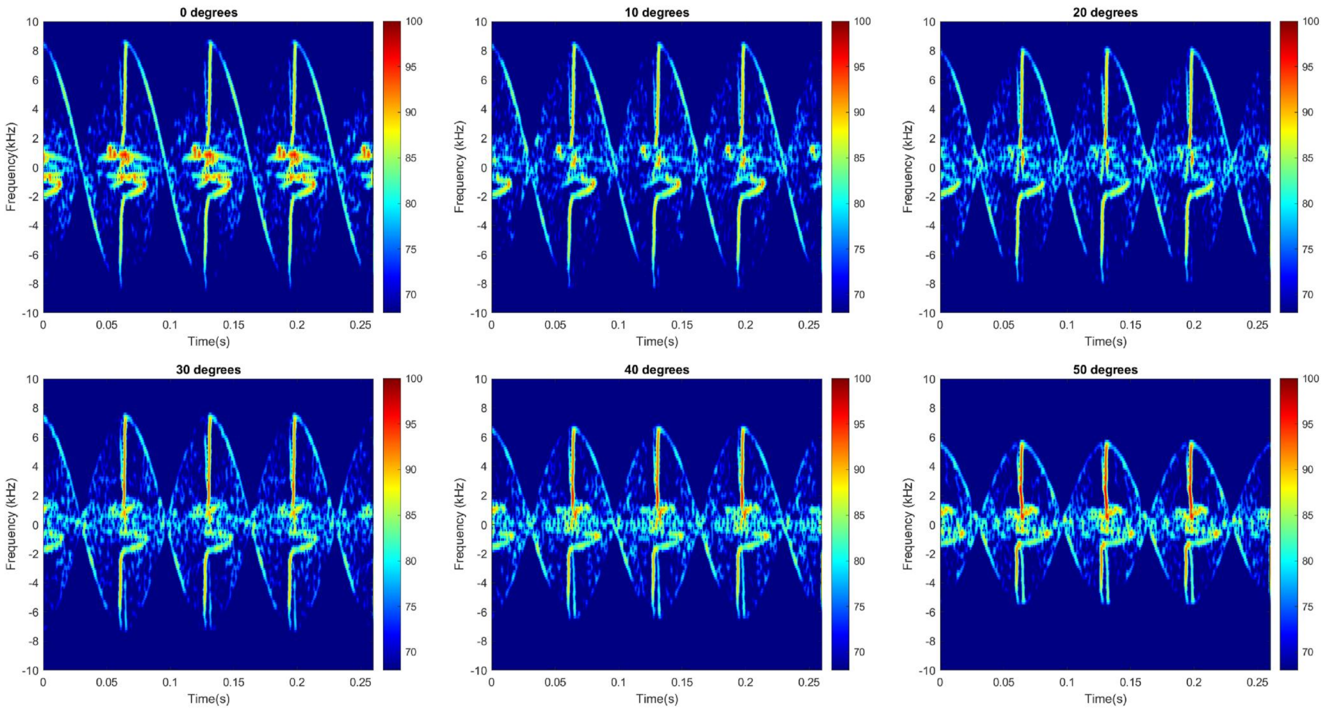

FIGURE 13 Simulated spectrograms showing micro-Doppler for the Joyance JT5L-404 for elevation angles ranging from 0 to 50°.

TABLE 1 Validation radar experimental parameters.

Parameter	Value
Propeller rotation frequency	<20 Hz
Maximum propeller length (tip-to-tip)	59 cm
Range to drone	3–5m

TABLE 2 Validation radar requirements.

Requirement	Value
CW operation	Required to fully sample micro-Doppler
High Doppler sensitivity and resolution	Capture high-fidelity Doppler features
Frequency	94 GHz
Max tip Doppler	19.7 kHz (3sf)
Antenna beamwidth (3 dB)	>11.2° (3sf)

required, although the radar can also operate in Frequency Modulated Continuous Wave mode.

Chirps centred around 1 GHz, are generated digitally in a Direct Digital Synthesiser, before being upconverted against a 6.833 GHz reference. They are then amplified and frequency doubled to 15.67 GHz. Copies of this signal are passed to the transmit and receive x6 frequency multiplier chains. Received signals are de-ramped in a homodyne I-Q mixer and the outputs are amplified and filtered before being sampled by an analog to digital converter (ADC).

For this application, there were some key design decisions to ensure the radar could produce high-fidelity micro-Doppler information. Firstly, the IF chain needed to handle the micro-Doppler features which were at very low IF frequencies, from DC to ±20 kHz. Wenzel LNAA-30-0 amplifiers were chosen for the IF amplification because of their flat response at very low frequencies, and low frequency roll-off [28]. Great care was taken to minimise spurious signals which might obscure the micro-Doppler propeller features. This involved using shielded cables to reduce pick-up, and extensively characterising all components to avoid those which introduced spurs, or using frequency regions without inherent spurs, wherever possible. Finally, smooth-walled feedhorn antennas, manufactured previously at the University of St Andrews, were used because of their 20° one-way 3 dB beamwidth, which was sufficient to fully contain the propellers at the ranges of interest. A final concern was maximising the dynamic range of the system. The quantisation noise of the system (a parameter of the ADC used) was −93.5 dBm (after FFT processing gain). The receiver noise floor (derived from thermal noise passed through the IF chain) was −123.5 dBm. This meant that the maximum achievable dynamic range of the system was limited by the IF gain. However, the system still provided sufficient signal to noise ratio and as was deemed an acceptable compromise considering the good performance of the IF amplifier. The quantisation and receiver noise floors along with the expected IF powers for several different RCS targets for the ranges of interest is shown in Figure 15.

The expected received power for the targets of interest is well above the noise floor at the measurement ranges, giving ample dynamic range for measurements.

4.2 | Radar testing

The radar was fully calibrated in amplitude and the IQ imbalance was measured and corrected. Any IQ imbalance in the system would produce asymmetries in the micro-Doppler produced, which would reduce the radar's ability to verify the simulation accurately. The radar was also Doppler calibrated. Tests were performed to ensure the Doppler information was correct using a target of known velocity. A basic optical encoder was designed and built to measure accurately the rotation rate of a propeller of known length. At the same time, the radar measured the micro-Doppler return of the propeller's motion. The spectrograms for each rotation rate were processed using a rectangular window to avoid spectral leakage reducing the frequency accuracy of the measurement. The expected blade-tip maximum Doppler was calculated for the rotation rates measured and compared with the result read off the spectrograms. This process was repeated for a metallic, square ended propeller and a plastic propeller and the results for both are shown in Figure 16.

For the metallic propeller, the results agree very well showing that the system is well calibrated in Doppler. The plastic propeller results agree for lower rotation rates, but the lower signal-to-noise and tapering tip shape from these propellers means that accurately extracting the maximum blade-tip Doppler for higher rotation rates is difficult, and the blade-tip Doppler is underestimated.

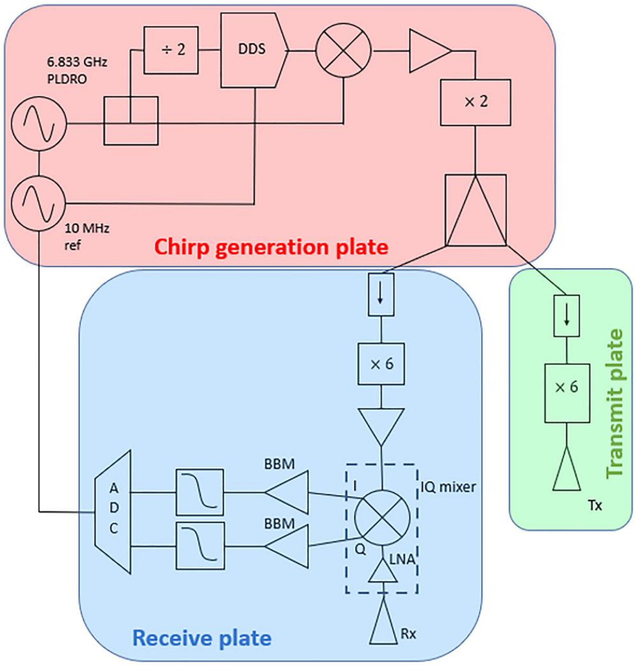

FIGURE 14 Simplified block diagram of validation radar.

4.3 | Radar physical design

The radar hardware is mounted on three metal plates. A chirp generating plate (where chirps are generated, upconverted, filtered and amplified with dual outputs at 15.67 GHz) and separate transmit and receive plates. The receive plate also contains the IF chain. The transmit and receive plates, which are quite portable and can be tripod mounted, are attached together for monostatic operation, or separated for bistatic measurements. The three plates are shown in Figure 17.

5 | SIMULATION VALIDATION

All three drones presented in Section 3.4 were measured experimentally to validate the simulation.

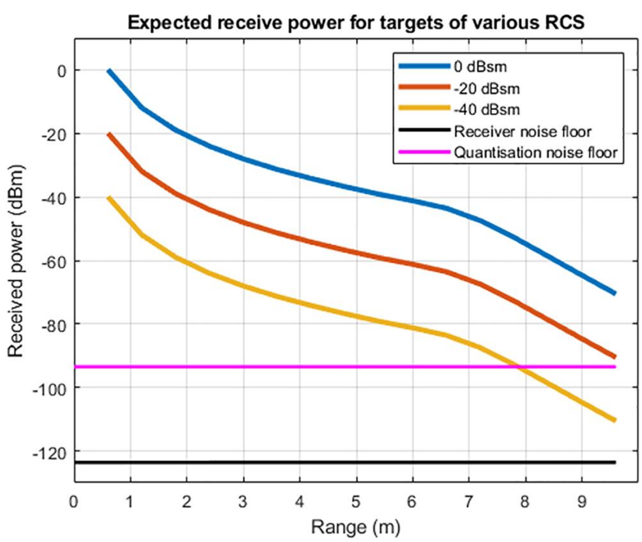

FIGURE 15 Expected receive power for typical radar cross-section (RCS) targets. NB: 0 dBsm ~ humans, −20 dBsm ~ drone fuselages and −40 dBsm ~ drone propellers.

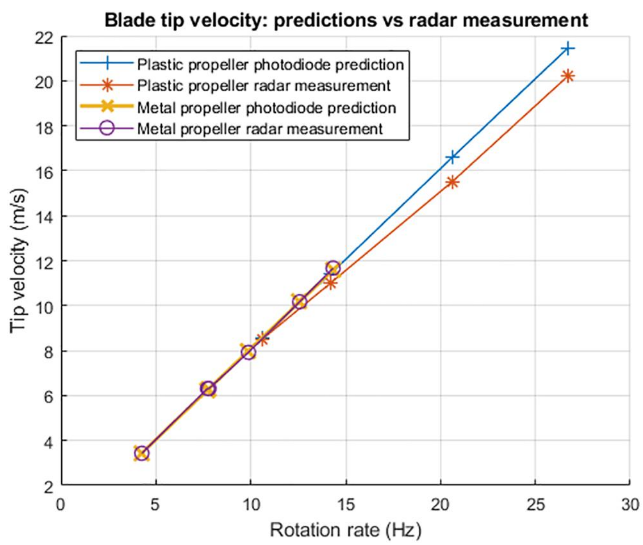

FIGURE 16 Radar measured blade-tip maximum Doppler versus predicted value from optical encoder.

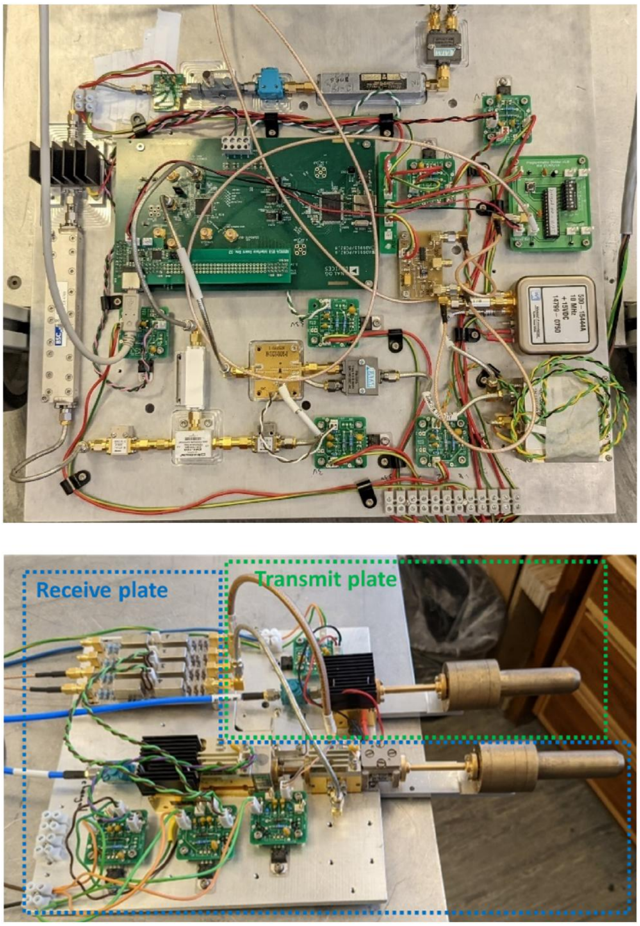

FIGURE 17 Chirp generating plate containing Direct Digital Synthesiser (DDS), reference signals and RF chain (top). Transmit and receive plates bolted together in monostatic operation (bottom). The feedhorn antennas are also shown.

5.1 | Validation setup

The simulated scenario was recreated as closely as possible for the purposes of validation. For the DJI S900 and the Joyance the power to all but one of the motors was disconnected such that only one propeller would rotate. For the DJI Phantom 3, it was not possible to disconnect the motor power, so three of the four propellers were removed. The transmit and receive plates of the radar were attached in the monostatic configuration (as shown in Figure 17) and the assembly was attached to a camera tripod. This allowed for portability of the radar and, combined with a digital inclinometer, convenient elevation angle control. A low-power laser diode was attached to the transmit feedhorn mount making it simple to align targets with antenna boresight. A section of the lab was covered in RF absorber and the drone placed in front. The setup is shown in Figure 18.

All three of the drones considered here can spin their propellers at an idling speed which is not sufficient to sustain flight. This varied between 7.5–20.0 Hz and meant that the propeller micro-Doppler could be measured in a controlled manner. For each drone, the single propeller micro-Doppler

FIGURE 18 Experimental setup for simulation validation.

signature was measured by the radar for elevation angles ranging from 0 to 60°.

5.2 | Results

The validation radar's DJI Phantom 3 micro-Doppler measurements for all elevation angles are shown in Figure 19 and the equivalent simulation results are shown in Figure 20. The experimental and simulation results appear to be in good agreement. The blade flash in both contains the distinctive S-shape which collapses in a similar fashion as the elevation angle increases. Furthermore, the asymmetry in the blade flash is alike for higher elevation angles with the positive Doppler return scattering more strongly than the negative Doppler return although this is more apparent in the simulation.

The sinusoidal blade tip trace appears similar with the falling edge around the apex containing the strongest return in experimental and simulated results. One notable difference is that the blade flash in the experimental results appears as a pair of less intense flashes which is only very weakly replicated in simulation. It is believed that these twin blade flashes are a result of the leading and trailing edges of the blade. As can be seen in Figure 4, these edges are not quite parallel and so would result in blade flashes at slightly different times. At 94 GHz the plastic propeller blade will be partially transparent, such that the validation radar is able to see through the blade resulting in scattering from the obscured edge. Alternatively, this could be an edge diffraction effect. Neither of these phenomena are handled by the simulation. The experimental and simulation results for the DJI S900 are shown in Figures 21 and 22 respectively.

The blade flash is slightly displaced laterally from the centre of the sinusoid in both figures. Finally, the blade tip sinusoidal trace appears comparable between the two results with both containing the stronger the falling edge. A slight difference between the results is the second much fainter blade flash seen experimentally which may be a result of the leading and trailing edge of the blade. Furthermore, experimentally the blade flash is slightly slanted or bent, an effect that increases with elevation angle which is not seen in simulation. This may be a result of the malleable plastic blade flexing as it rotates. The experimental and simulated results for the Joyance JT5L-404 are shown in Figures 23 and 24. The low frequency region has

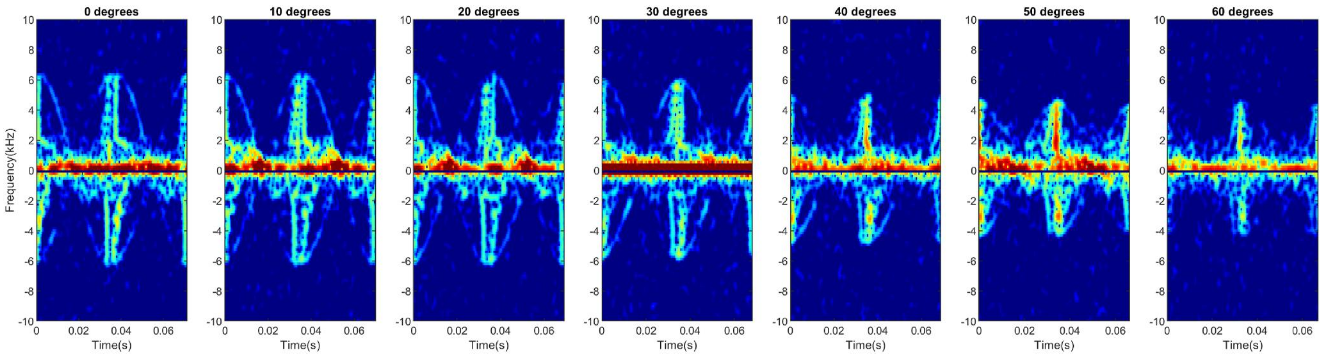

FIGURE 19 Experimental micro-Doppler signatures for Phantom 3 for elevation angles from 0 to 60°.

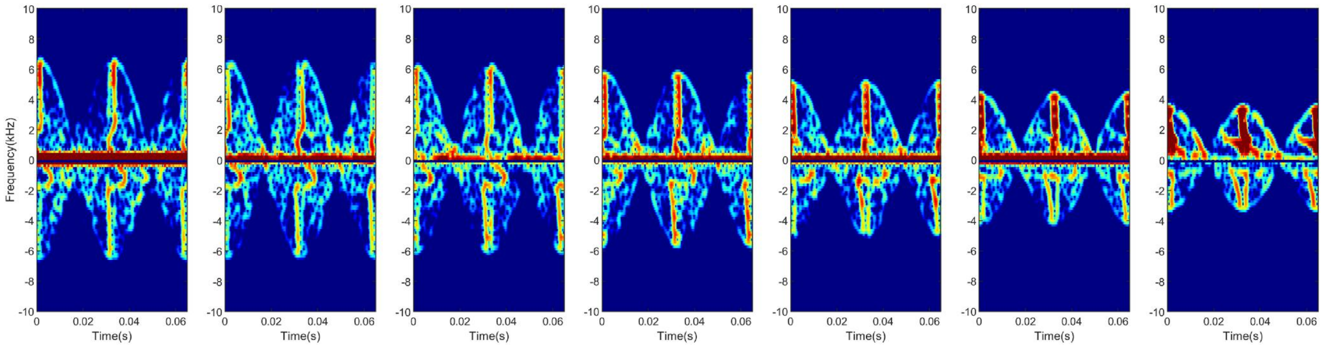

FIGURE 20 Simulated micro-Doppler signatures for Phantom 3 for elevation angles from 0 to 60°.

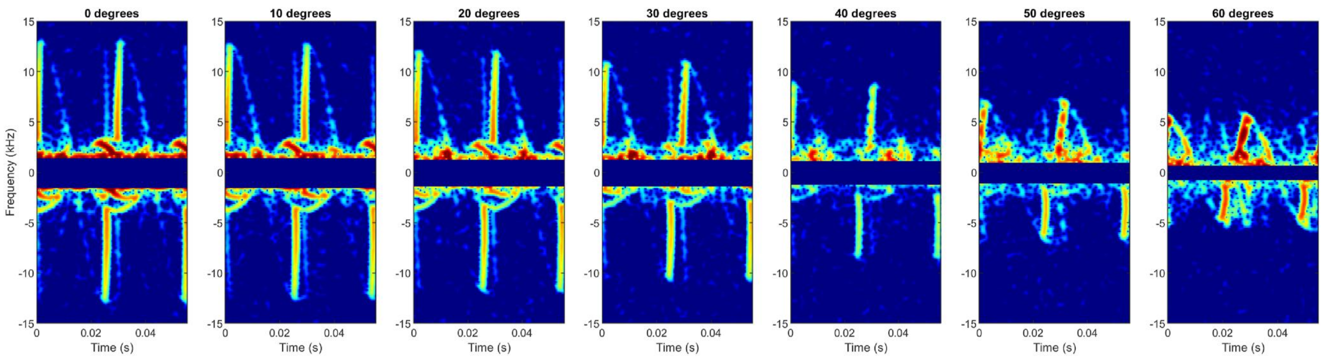

FIGURE 21 Experimental micro-Doppler for DJI S900 for elevation angles from 0 to 60°.

been filtered out for the same reasons as above. The blade flash for the experimental and simulated results appear similar as a long thin line tapering down to a point. In both results, the negative Doppler portion of the blade flash exhibits an upside-down L-shaped feature, with the bend angle of this increasing with elevation angle. The blade tip trace is alike in both results, with the strongest return in the falling edge of the sinusoid. One difference is the multiple fainter blade flashes seen in the experimental result. The Joyance vibrated significantly when idling its propeller and it is believed this resulted in the multiple fainter blade flashes. The slant in the blade flash is also seen experimentally, but not in simulation.

The simulation efficacy depends significantly on the quality of the input 3D model. Using lower quality 3D scanners (such as the optical scanners shown in Section 3.3) results in much noisier spectrograms with significantly less detailed micro-Doppler features as shown in Figure 6. Moreover, using CAD models sourced from online libraries that do not perfectly match the dimensions of the real propeller produce micro-Doppler features that only weakly agree with experimental results. For example, a DJI S900 propeller downloaded from the online CAD library GrabCAD is shown in Figure 25 (left) alongside its real equivalent in Figure 25 (right).

The propellers appear visually similar (although some differences are easily seen), but when compared with the laser scanned point cloud, significant discrepancies between the two emerge. The simulated spectrogram using the downloaded S900 CAD model for an elevation angle of 0° (Figure 26) is

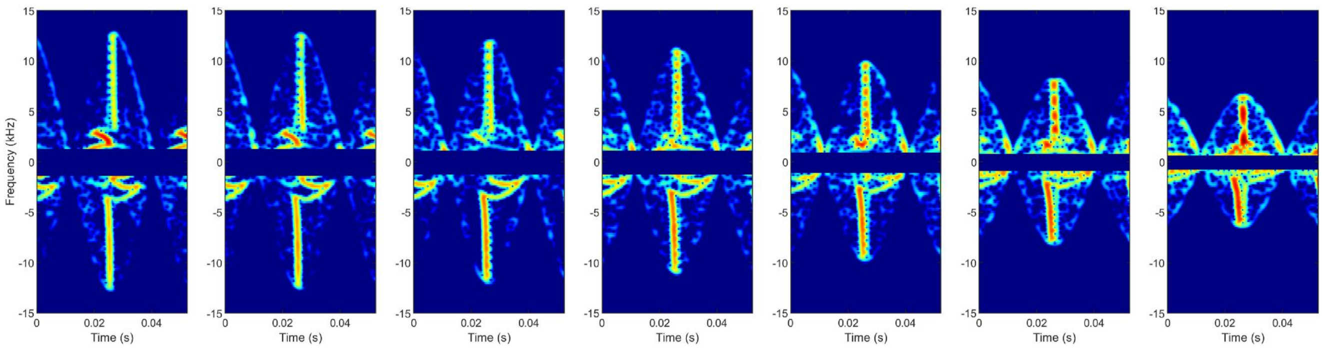

FIGURE 22 Simulated micro-Doppler for DJI S900 for elevation angles from 0 to 60°. Note that the lowest frequencies have been filtered out in both to maximise equivalence between the simulated and experimental scenarios. This is because the experimental result contains the full drone whereas the simulation result only contains the propeller. The experimental and simulation results appear to be in good agreement. The blade flash in both is of a similar shape appearing as a long thin stripe which reduces in extent as the elevation angle increases. The reverse S-shape feature at the base of the blade flash is alike in the experimental and simulated results, as is its collapse with increasing elevation angle. Both results contain a curved feature that links the top and bottom of the reverse S feature to the main blade flash, and it is especially prominent in the negative Doppler for both.

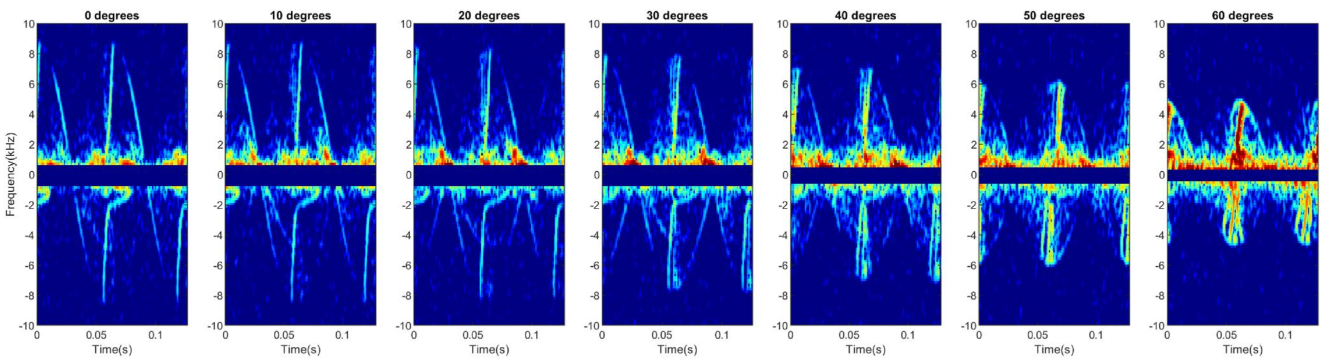

FIGURE 23 Experimental micro-Doppler for Joyance JT5L-404 for elevation angles from 0 to 60°.

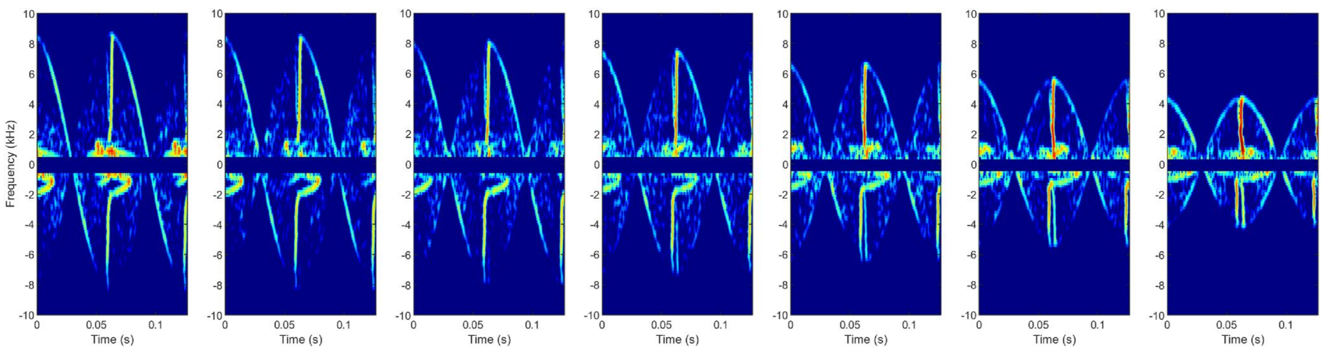

FIGURE 24 Simulated micro-Doppler for Joyance JT5L-404 for elevation angles from 0 to 60°.

very different from both the simulated result using the laser scanned model and the experimental result.

Using laser scanned components offer a good alternative to downloaded CAD models that ensures reliable results.

5.3 | Quantitative analysis

The similarity between the simulated and experimental spectrograms was measured quantitatively, via a normalised cross correlation. Initial results were limited by the curvature of the blade flash seen experimentally. The normalised cross correlation compares the agreement on a pixel-by-pixel basis; consequently, the slight bending in the blade flash meant that the overlap between the experimental and simulated figures was quite low. Figure 27 (left) shows example simulated and experimental results overlaid for the DJI S900.

To correct for the bending, a horizontal pixel shift was applied to the experimental spectrograms in proportion to the vertical distance of the pixel from the zero Doppler horizontal. An example spectrogram for the DJI S900 before and after this rotation shift operation is shown in Figure 28.

FIGURE 25 Downloaded CAD model of DJI S900 (left). Real DJI S900 propeller (right). CAD, computer-aided design.

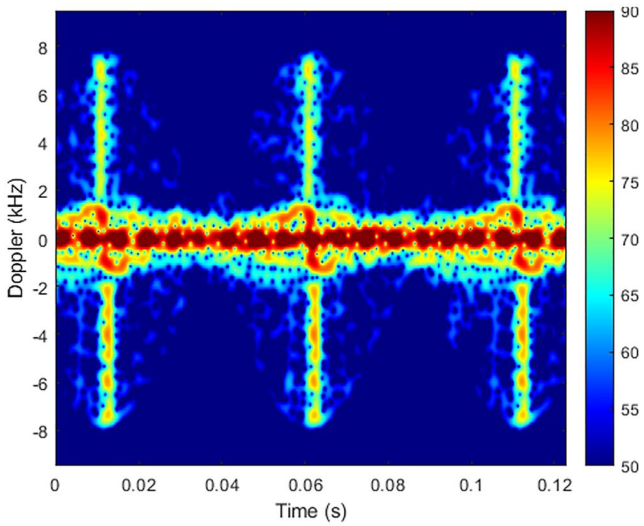

FIGURE 26 DJI S900 simulated micro-Doppler using CAD model downloaded from GrabCAD for 0° elevation angle. CAD, computer-aided design.

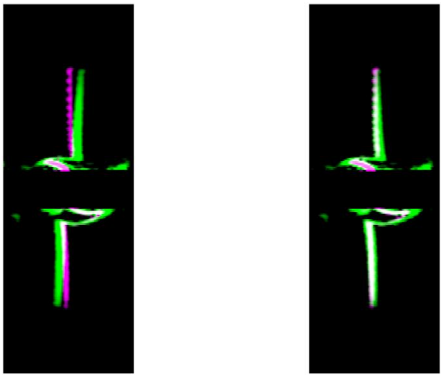

FIGURE 27 Overlap between experimental DJI S900 measurement (green) and simulated DJI S900 result (purple) before (left) and after (right) bend correction.

Figure 27 (right) shows the simulated result and experimental result after the bend correction for the DJI S900 overlaid, demonstrating a clear improvement in agreement over Figure 27 (left).

The normalised cross correlation (after the bend correction) was taken between the experimental result for zero degree elevation angle (referred to as in-plane), and the six simulated results for successive elevation angles from 0–50°. The results of this, for the three drones used, is given in Figure 29. A value of one would indicate perfect agreement between the simulation and experiment. Note that the low Doppler region was removed for all three drones in order to ensure an equivalent comparison could be made between the results.

Figure 29 shows that the normalised cross-correlation performance of the simulation is very different for laser scanned models and models sourced from online libraries. For the laser scanned models (DJI S900 and Joyance JTL5L-404), the normalised cross correlation for the in-plane comparison is 0.7–0.8 and decreases as simulated elevation angle increases as expected. The absolute agreement between the in-plane simulated and experimental results represents good, but not exceptional agreement. The simulated and experimental (post bend correction) in-plane results are plotted on top of one another in Figure 27 (right). Although the blade flash and tip trace are in good agreement, there are a number of experimental features (shown in green), that are not replicated in the simulation. Most of these features are contained at lower Doppler frequencies and may correspond to drone components which are not simulated (fuselage, motors etc.). Consequently, it should be expected that perfect cross-correlation will not be achieved here and that a result in excess of 0.7 or 70% represents good agreement. In contrast, the DJI Phantom 3 model, obtained from online libraries, demonstrated significantly inferior normalised cross-correlation performance, remaining consistently low across all elevation angles. This shows that the simulation performance is dependent on using very accurate input models, which can be achieved using a laser scanner.

To demonstrate the simulation's capability to generate sufficiently accurate results for internal classification purposes, the authors conducted transfer learning experiments utilising simulated data. SqueezeNet, a pre-trained convolutional neural network generated by the University of California, Berkley and Stanford was trained on exclusively simulated data for all elevation angles for the three drones considered [29]. 90 images were included for each class. The network was tested on (unseen) experimental data for in-plane measurements only, and it achieved 100% classification accuracy for all three drone classes.

5.4 | Analysis

The experimental results from the validation radar are in good agreement with the simulated results especially for propellers that were laser scanned. The simulation can accurately replicate subtle and unique micro-Doppler features with high levels of detail using only a CAD model or laser scanned point cloud. Blade flash and sinusoidal blade tip trace are recreated accurately with similar relative scattering strengths. The simulation is able to generate synthetic data which contain the micro-

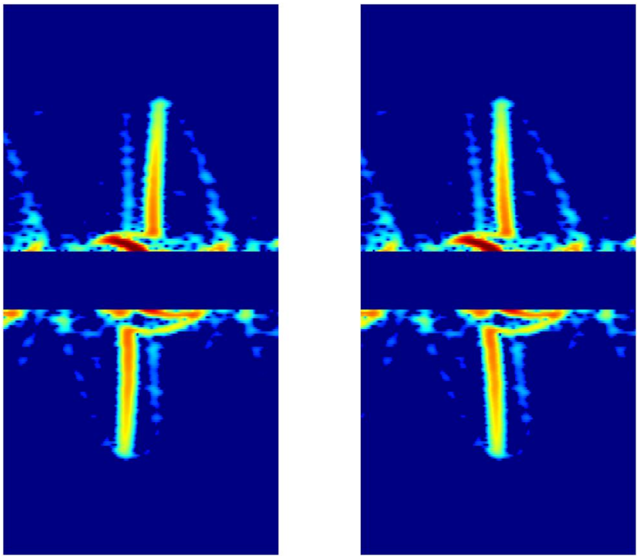

FIGURE 28 DJI S900 result before bend correction (left) and afterwards (right).

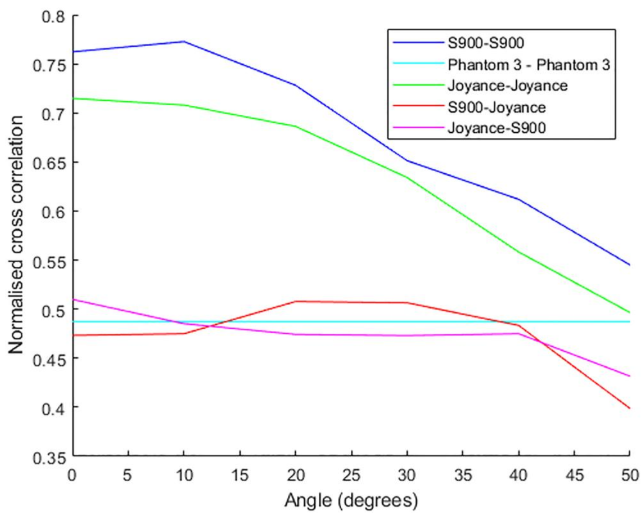

FIGURE 29 Normalised cross correlation between in-plane experimental result and simulated result for elevation angles 0–50°. Note: first drone in the legend corresponds to the experimental result and the second the simulated.

Doppler features seen in experimental data over a wide range of elevation angles. Furthermore, this data is sufficiently accurate that it can be used to train a neural network applied to classifying experimental data effectively. In addition, we have found the simulation methodology to be very sensitive to the accuracy of the CAD model used, and laser scanned models have been shown to generate the best results. Although only single propeller examples have been considered in this paper, the simulation methodology presented here can be trivially extended to include multiple propellers (with increased computational effort). Going forwards, the simulation will be used to generate micro-Doppler returns from full drones. These datasets can then be used to train classifiers for use on in-flight experimental data.

6 | CONCLUSION

In this paper, a micro-Doppler simulation has been presented capable of generating detailed and accurate micro-Doppler spectrograms from 3D models, after converting them to point clouds. The entire workflow of this simulation, from 3D model to spectrogram has been described including the process of sourcing 3D files of drone components. A 94 GHz radar has been designed and built for validating this simulation. This paper elucidates the specific design decisions that were made for the radar system, as well as the testing and calibration processes employed to ensure the generation of high-fidelity micro-Doppler spectra. Furthermore, the physical design has been given. Experimental data, captured from a setup designed to replicate the simulation as closely as possible, has been presented for comparison with the simulated results. Very good agreement between equivalent simulated and experimental results has been demonstrated. Features such as blade flash and sinusoidal blade-tip trace seen experimentally have been replicated in the simulation with very high detail and accuracy, including relative scattering strength. The normalised cross correlation between the simulated and experimental results exceeded 70% for laser scanned components, which is considered a good result due to the presence of micro-Doppler features in the experimental result from components that are not simulated. This is backed up by good transfer learning performance using synthesised data. Whilst the simulation has been validated at 94 GHz the methodology is applicable at other frequencies. Using only a 3D scan as an input, the simulation is able to accurately replicate the unique micro-Doppler features present in diverse propeller returns. Additionally, its computational efficiency compared to EM solvers at millimetre-wave frequencies enables the generation of large amounts of synthetic data over a wide range of elevation angles quickly and easily. Ultimately, the datasets generated by this simulation can be used to train neural networks to perform internal classification.

AUTHOR CONTRIBUTIONS

Matthew Moore developed and implemented the simulation methodology presented in this paper. He also developed, built, and tested the radar and took the data that was used to validate the simulation including all the signal processing. MM wrote the paper. Duncan Robertson and Samiur Rahman are the supervisors for Matthew Moore's PhD. DR secured the funding for Matthew Moore's PhD. Duncan Robertson supervised the development of the radar and SR supervised the development of the simulation and data acquisition code. They both edited the paper.

ACKNOWLEDGEMENTS

The authors thank colleagues Rob Hunter for his help with the CAD models, Ryan Milne for the 3D scanning and Adam Robertson for his work during his master's project. We acknowledge the financial support of the Engineering and Physical Sciences Council and QinetiQ who are funding MM's PhD.

CONFLICT OF INTEREST STATEMENT

We have no conflicts of interest to disclose.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available at https://doi.org/10.17630/9e463540-c7b5-4cdf-a10c-cebcc4659eec.

ORCID

Matthew Moore https://orcid.org/0000-0002-5016-7736
Samiur Rahman https://orcid.org/0000-0002-5477-4218

REFERENCES

1. Skies without Limits v2.0 - PwC UK. https://www.pwc.co.uk/issues/emerging-technologies/drones/the-impact-of-drones-on-the-uk-economy.html. Accessed 3 February 2023
2. Police to Crack Down on Drones Flown Dangerously - BBC News. https://www.bbc.co.uk/news/technology-57512513. Accessed 3 February 2023
3. How Dangerous are Drones to aircraft? | Air Transport | the Guardian. https://www.theguardian.com/technology/2018/dec/20/how-dangerous-are-drones-to-aircraft. Accessed 3 February 2023
4. Warning over Drones use by Terrorists - BBC News. (2016). https://www.bbc.co.uk/news/technology-35280402. Accessed 15 April 2021
5. Harmanny, R.I.A., De Wit, J.J.M., Premel-Cabic, G.: Radar micro-Doppler mini-UAV classification using spectrograms and cepstrograms. Int. J. Microw. Wirel. Technol. 7(3–4), 469–477 (2015). https://doi.org/10.1017/s1759078715001002
6. Chen, V.C.: The Micro-Doppler Effect in Radar, ed. Artech House (2019)
7. Ritchie, M., et al.: Monostatic and bistatic radar measurements of birds and micro-drone. In: 2016 IEEE Radar Conf. RadarConf 2016, pp. 2–6 (2016)
8. De Wit, J.J.M., Harmanny, R.I.A., Molchanov, P.: Radar micro-Doppler feature extraction using the Singular Value Decomposition. In: 2014 Int. Radar Conf. Radar 2014, pp. 1–6 (2014)
9. Torvik, B., Olsen, K.E., Griffiths, H.: Classification of birds and UAVs based on radar polarimetry. IEEE Geosci. Remote Sens. Lett. 13(9), 1305–1309 (2016). https://doi.org/10.1109/lgrs.2016.2582538
10. Rahman, S., Robertson, D.A.: In-flight RCS measurements of drones and birds at K-band and W-band. IET Radar, Sonar Navig. 13(2), 300–309 (2019). https://doi.org/10.1049/iet-rsn.2018.5122
11. Dale, H., et al.: A comparison of convolutional neural networks for low SNR radar classification of drones. IEEE Natl. Radar Conf. - Proc., 1–5 (2021)
12. Klaer, P., et al.: An investigation of rotary drone herm line spectrum under manoeuvering conditions. Sensors 20(20), 1–15 (2020). https://doi.org/10.3390/s20205940
13. Rahman, S., Robertson, D.A.: Millimeter-wave micro-Doppler measurements of small UAVs. Radar Sens. Technol. XXI 10188(May 2017), 101880T (2017)
14. Rahman, S., Robertson, D.A.: Multiple drone classification using millimeter-wave CW radar micro-Doppler data. Radar Sens. Technol. XXIV 1140809(April 2020), 653–661 (2020). https://doi.org/10.1049/iet-rsn.2019.0493
15. Bjorklund, S., Wadstromer, N.: Target detection and classification of small drones by deep learning on radar micro-Doppler. 2019 Int. Radar Conf. RADAR 2019, 182–185 (2019)
16. Schröder, A., et al.: Numerical RCS and micro-Doppler investigations of a consumer UAV. In: Stein, K.U., Schleijpen, R.H.M.A. (eds.) Target Backgr. Signatures II 9997 (2016). 999704. October 2016
17. Speirs, P.J., et al.: Comparisons between simulated and measured X-band signatures of Quad-, Hexa- and Octocopters. In: 2018 15th Eur. Radar Conf. EuRAD 2018, pp. 325–328 (2018)
18. Chen, V.C., et al.: Micro-Doppler effect in radar: phenomenon, model, and simulation study. IEEE Trans. Aerosp. Electron. Syst. 42(1), 2–21 (2006). https://doi.org/10.1109/taes.2006.1603402
19. Martin, J., Mulgrew, B.: Analysis of the theoretical radar return signal from aircraft propeller blades. In: IEEE Int. Conf. Radar, pp. 569–572. Publ by IEEE, Arlington (1990)
20. Melino, R., Bourne, C., Tran, H.T.: Modelling Helicopter Radar Backscatter (2011)
21. Cai, Y., Krasnov, O., Yarovoy, A.: Simulation of radar micro-Doppler patterns for multi-propeller drones. In: 2019 Int. Radar Conf. RADAR 2019, pp. 1–5 (2019)
22. Katz, S., Tal, A., Basri, R.: Direct visibility of point sets. ACM Trans. Graph. 26(3), 24–es (2007). https://doi.org/10.1145/1276377.1276407
23. CloudCompare. https://www.danielgm.net/cc/
24. GrabCAD: Design Community, CAD Library, 3D Printing Software. https://grabcad.com/. Accessed 18 May 2021
25. EinScan-SE Powerful Desktop 3D Scanner | EinScan. https://www.einscan.com/einscan-se/. Accessed 3 February 2023
26. Laser Scanning | 3D Scanning, Reverse Engineering, Inspection Reports. https://www.laser-scanning.co.uk/. Accessed 3 February 2023
27. Moore, M., Robertson, D.A., Rahman, S.: 94 GHz Doppler radar for experimental validation of small UAV micro-Doppler. Proc. SPIE 12108. Radar Sens. Technol. XXVI, 121080S 12108, 281–291 (2022). SPIE
28. Audio Amplifier (LNAA) - Wenzel Associates, Inc. https://wenzel.com/model/audio-amplifier-lnaa/. Accessed 3 February 2023
29. Iandola, F.N., et al.: SqueezeNet: AlexNet-Level Accuracy with 50x Fewer Parameters and <0.5MB Model Size. ICLR 2017 (2017)

How to cite this article: Moore, M., Robertson, D.A., Rahman, S.: A new simulation methodology for generating accurate drone micro-Doppler with experimental validation. IET Radar Sonar Navig. 1–16 (2023). https://doi.org/10.1049/rsn2.12494